

PRACTICAL REPORT 05

FULL STACK DEVELOPMENT (FSD) — SEMESTER 5

Topic: Bootstrap 5 Integration & Registration UI
Target: Online CDN, Offline Assets & Form Design

Practical Objectives & Tasks

1. **Task I:** Demonstrate the implementation of Bootstrap online (using the CDN link).
2. **Task II:** Demonstrate the implementation of Bootstrap offline (using local stylesheet and script files).
3. **Task III:** Create a simple web page using Bootstrap 5 with HTML for student registration including page title, roll number, first name, mobile number, email ID, and address.

 **Architecture & Implementation Strategy:** Our primary Campus Companion repository utilizes modular Vanilla CSS for granular aesthetic control. To fulfill academic requirements without polluting the production React codebase, standalone self-contained implementations for Practical 5 were created and tested under `reports/practical_05/`.

Theoretical Foundation & Key Concepts

1. Content Delivery Network (CDN) vs Self-Hosted Local Integration

Online CDN (Content Delivery Network): Assets are delivered from geographically distributed edge servers (such as jsDelivr or Cloudflare). Provides fast global delivery, browser cache sharing across sites, and zero local disk footprint, with Subresource Integrity (SRI) for security.

Offline Local Integration: Assets (`bootstrap.min.css` and `bootstrap.bundle.min.js`) are hosted directly within the application directory. Crucial for intranet, lab environments, enterprise air-gapped networks, and environments requiring strict version immutability.

2. Bootstrap 5 Form Architecture & Validation Framework

Bootstrap 5 provides semantic CSS form controls (`.form-control`, `.form-label`, `.form-select`, `.input-group`) alongside a pseudo-class based validation system. By adding `novalidate` and toggling `.was-validated` via JavaScript, form fields dynamically display green/red borders with `.valid-feedback` and `.invalid-feedback` hints.

TASK I Bootstrap 5 Online Implementation (CDN Integration)

File: `reports/practical_05/task1_bootstrap_online.html`

```
<!-- Bootstrap 5 CSS via jsDelivr CDN with SRI -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
      rel="stylesheet"
      integrity="sha384-QWTKZyjpPEjISv5WaRU90FeRpok6YctnYmDr5pNlyT2bRjXh0JMhJY6hW+ALEwIH"
      crossorigin="anonymous">

<!-- Bootstrap Icons CDN -->
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.min.css">

<!-- Interactive 12-Column Responsive Grid Layout -->
<div class="container my-4">
  <div class="row g-4">
    <div class="col-md-4">
      <div class="card h-100 shadow-sm">
        <div class="card-body text-center">
          <h4 class="card-title fw-bold">Zero Setup</h4>
          <p class="card-text text-muted">Instant inclusion via remote CDN links.</p>
        </div>
      </div>
    </div>
  </div>
</div>

<!-- Bootstrap 5 JavaScript Bundle (Popper Included) -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
        integrity="sha384-YvpcrYf0tY3LHB60NNkmXc5s9fDVZLESaAA55NDz0XhY9GkcIdslK1eN7N6jIeHz"
        crossorigin="anonymous"></script>
```

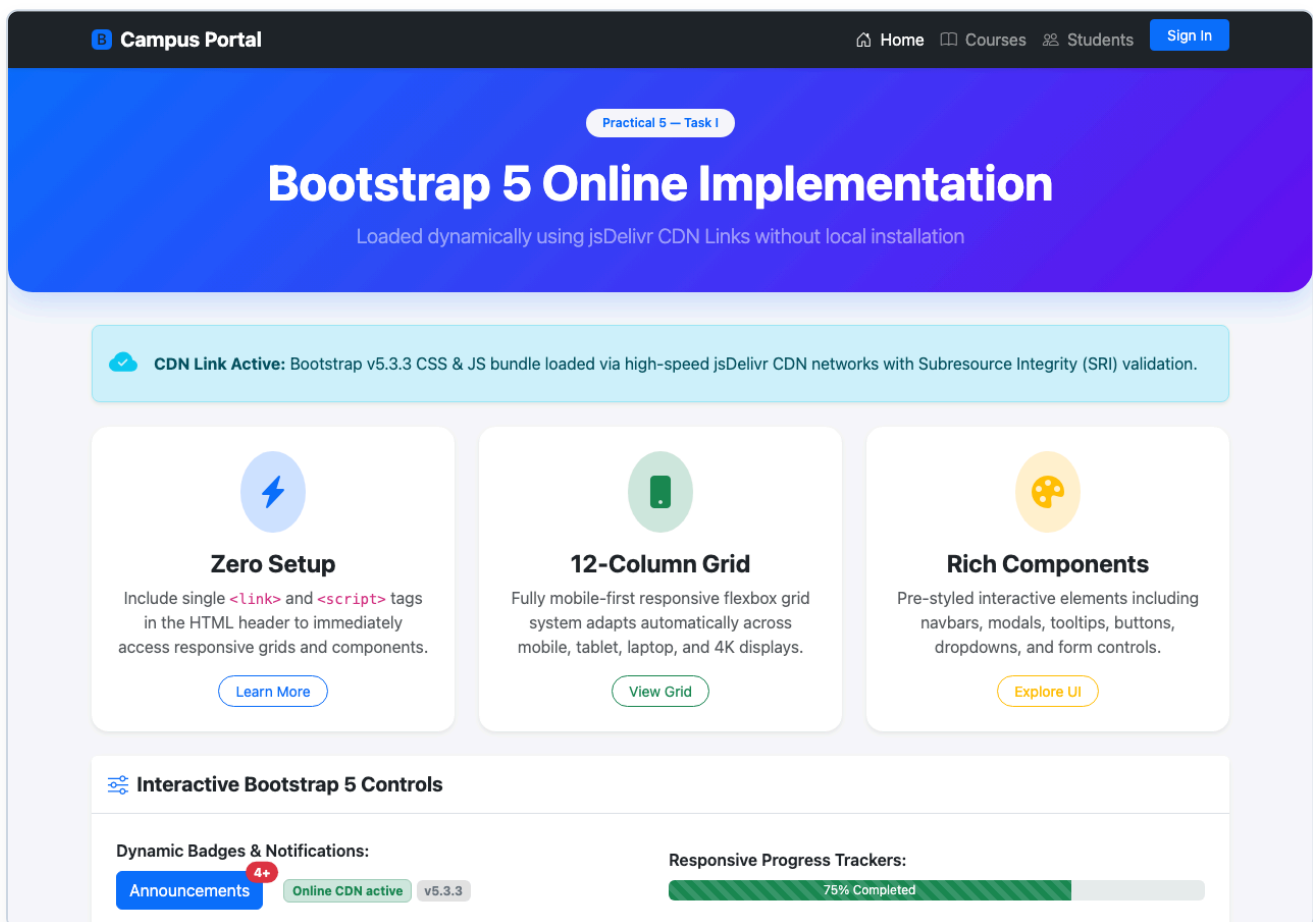


Figure 5.1: Bootstrap 5 loaded and rendered online via jsDelivr CDN links.

TASK II Bootstrap 5 Offline Implementation (Local Static Assets)

File: reports/practical_05/task2_bootstrap_offline.html | Assets: reports/practical_05/assets/

```
<!-- Offline Local Bootstrap 5 CSS (Self-Hosted) -->
<link rel="stylesheet" href="./assets/css/bootstrap.min.css">

<!-- Offline Components: Accordions & Responsive Tables -->
<div class="accordion" id="offlineAccordion">
  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button" type="button" data-bs-toggle="collapse" data-bs-target="#collapseOne">
        1. Intranet & Air-Gapped Environments
      </button>
    </h2>
    <div id="collapseOne" class="accordion-collapse collapse show" data-bs-parent="#offlineAccordion">
      <div class="accordion-body">
        Offline assets guarantee full functionality without external internet connection.
      </div>
    </div>
  </div>
</div>

<!-- Offline Local Bootstrap 5 JavaScript Bundle -->
<script src="./assets/js/bootstrap.bundle.min.js"></script>
```

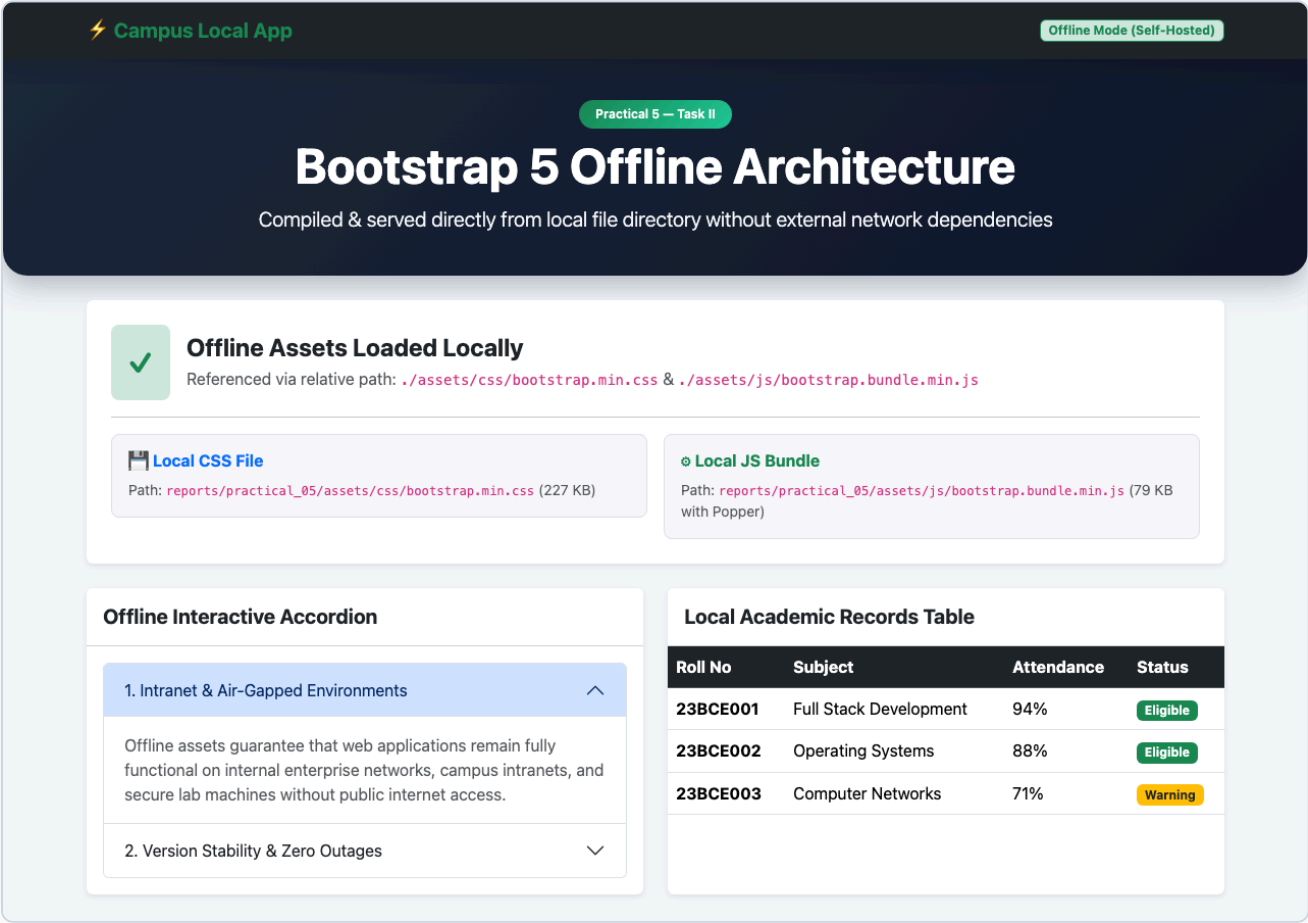


Figure 5.2: Bootstrap 5 running completely offline with local minified CSS and JS assets.

TASK III Bootstrap 5 Student Registration Web Page

File: reports/practical_05/task3_student_registration.html (Includes all mandatory criteria)

```

<!-- Page Title, Roll Number, First Name, Mobile Number, Email ID, Address -->
<div class="card card-custom">
  <div class="form-header">
    <h2 class="fw-bold mb-1">Student Registration Form</h2>
    <p class="mb-0 text-white-50">Academic Admission Portal</p>
  </div>
  <div class="card-body p-4 p-md-5">
    <form id="studentForm" novalidate>
      <!-- Roll Number -->
      <div class="col-md-6">
        <label for="rollNumber" class="form-label">Roll Number / Enrollment No *</label>
        <input type="text" class="form-control" id="rollNumber" placeholder="e.g. 23BCE001" required>
      </div>
      <!-- First Name -->
      <div class="col-md-6">
        <label for="firstName" class="form-label">First Name *</label>
        <input type="text" class="form-control" id="firstName" placeholder="e.g. Shrey" required minlength="2">
      </div>
      <!-- Mobile Number -->
      <div class="col-md-6">
        <label for="mobileNumber" class="form-label">Mobile Number *</label>
        <input type="tel" class="form-control" id="mobileNumber" placeholder="9876543210" pattern="[6-9][0-9]{9}" required>
      </div>
      <!-- Email ID -->
      <div class="col-md-6">
        <label for="emailId" class="form-label">Email ID *</label>
        <input type="email" class="form-control" id="emailId" placeholder="shrey@campus.edu" required>
      </div>
      <!-- Residential Address -->
      <div class="col-12">
        <label for="address" class="form-label">Full Address *</label>
        <textarea class="form-control" id="address" rows="3" placeholder="Street, City, State..." required></tex>
      </div>
      <button type="submit" class="btn btn-primary">Submit Registration</button>
    </form>
  </div>
</div>

```

Campus Companion

FSD Practical 05

Student Admission 2025-2026

Student Registration Form

Please fill in your academic, contact, and residential details

1. Academic Identity

Roll Number / Enrollment No *

First Name *

e.g. 23BCE001

e.g. Shrey

2. Contact Details

Mobile Number *

Email ID *

e.g. 9876543210

e.g. shrey.patel@campus.edu

3. Residential Address

Full Address *

Enter complete house number, street, city, state, and pin code

Reset

Submit Registration

Figure 5.3: Bootstrap 5 Responsive Student Registration Web Page with validation and responsive grid.

Comparative Analysis: CDN vs Local Hosting

Evaluation Parameter	Online Bootstrap (CDN)	Offline Bootstrap (Local Files)
Setup Complexity	Minimal (single <link> and <script>)	Requires downloading and managing CSS/JS folders
Network Dependency	Requires active internet connectivity to load	Works completely offline & in air-gapped intranets
Caching & Latency	Edge CDN caching across global servers	Served directly from local server origin
Version Control	Controlled by URL version tag (e.g., v5.3.3)	Committed and locked in repository build

file:///Users/patelshreymukeshbhai/Desktop/College/sem5/FSD/Campus-Companion/reports/practical_05/Practical_05_Report.html

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✓ Practical Summary & Conclusion

All objectives of Practical 05 were successfully implemented, verified, and captured:

- **Online CDN Deployment:** Successfully demonstrated Bootstrap 5 integration with modern responsive navbar, grid cards, badges, and progress bars.
- **Offline Architecture:** Self-hosted compiled `bootstrap.min.css` and `bootstrap.bundle.min.js` locally, ensuring offline resilience.
- **Student Registration Interface:** Created a responsive, accessible, and validated form containing the Page Title, Roll Number, First Name, Mobile Number, Email ID, and Full Address.